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Dynamic orthotic device

Abstract

A dynamic orthotic device includes upper and lower members which rotate relative to one another allowing the heel of the wearer's foot to engage a fresh, different portion of the device, gently messaging the contacting portion of the wearer's heel greatly reducing the likelihood of plantar faciitis and other foot related problems caused by constant pressure resulting from continuous standing or walking.

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Background/Summary

BACKGROUND AND SUMMARY OF THE INVENTION

(1) The present invention is directed to the field of orthotics. More particularly, the present invention is directed to a dynamic orthotic device.

(2) As a rule, orthotic devices are static, that is, there is no significant movement, with some exceptions in cases where the device itself is flexible and, accordingly, the heel/foot sees the same pressure in the same place with each successive step. This means that the exertion of pressure on the same spot can produce a “pressure overload” leading/contributing to plantar faciitis and other related foot problems which, typically, orthotics nominally, attempt to remedy.

(3) It is among the objects of this invention to provide the first truly dynamic orthotic device in which the pressure administration point (“pressure point”) changes with each step thereby eliminating the pressure buildup that can take place with walking or other forms of exercise. Indeed, with the current dynamic orthotic device, a slight heel-toe rocking motion while standing can produce the desired change in the pressure point providing a “foot massage” as the heel-toe rocking motion activates the dynamic orthotic device.

(4) The present invention is comprised of a dynamic orthotic device for massaging a heel portion of the sole of a wearer's foot, the dynamic orthotic device which includes a) a base member; b) means to house a plurality of spherical balls beneath the heel portion of the sole of the wearer's foot; c) a first means to rotate said means to house in a first rotational direction. The means to house a plurality of spherical balls can include a first rotating disk having a first plurality of spherical ball-receiving holes and a second rotating disk having a second plurality of spherical ball-receiving holes. A second means to rotate the means to house can be included wherein the first means to

rotate rotates the first rotating disk in a first rotational direction and the second means to rotate rotates the second rotating disk in a second opposite rotational direction. Each of the first and second means to rotate may each include a gear-toothed member associated with its respective rotating disk and a plurality of spring-biased arms which engage the gear-toothed member. In addition, the device can include a spring-biased pushbutton engaged by a heel of the wearer, the spring-biased pushbutton mounting the pluralities of the spring-biased arms.

(5) In a second embodiment, the dynamic orthotic device for massaging a heel portion of the sole of a wearer's foot includes: a) a first upper member; b) a second lower member supporting the first upper member; c) spring means to bias the first member away from the second member; d) rotation means for turning the first member relative to the second member in response to the upper member being weighted and unweighted by the wearer's foot. In this embodiment, the upper member comprises a first outer cylinder having a flange protruding outwardly from a lower edge thereof and a second inner cylinder, the first outer cylinder and the second inner cylinder being interconnected by an upper surface. The second lower member comprises a first outer cylinder having an outwardly protruding flange extending outwardly from a lower edge portion of the first outer cylinder and a second inner cylinder, the first and second cylinders of the second lower member being interconnected by a bottom surface and interdigitating with the first and second cylinders of the first upper member. The second lower member further comprises a first set of upwardly directed, forwardly slanting teeth opposed by a second offset set of downwardly directed forwardly slanting teeth each secured to an inner wall of the first outer cylinder of said second lower member.

(6) The rotation means for turning said first upper member relative to the second lower member comprises a plurality of pins extending outwardly from said second inner wall of the first upper member which engage the first and second sets of teeth secured to said inner wall of the first outer cylinder of the second lower member, whereby when the upper surface of the first upper member is engaged by a heel portion of the sole of a wearer's foot, the upper member collapses against a force of the spring causing the plurality of pins to engage the upwardly directed forwardly slanting teeth of the second lower member and the plurality of pins to engage the downwardly directed forwardly slanting teeth when the heel portion of the sole of the wearer's foot is lifted from the upper surface of the upper member. The upper surface of the first upper member further comprises a plurality of massaging nodules secured to the upper surface.

(7) Various other features, advantages, and characteristics of the present invention will become apparent after a reading of the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The preferred embodiment(s) of the present invention is/are described in conjunction with the associated drawings in which like features are indicated with like reference numerals and in which

(2) FIG. 1 is a top view of a first embodiment of the dynamic orthotic device of the present invention;

(3) FIG. 2 is a front perspective view of the first embodiment;

(4) FIG. 3 is a cross-sectional side view of the first embodiment;

(5) FIG. 4 an exploded front perspective view of the first embodiment;

(6) FIG. 5 is a front perspective view of a second embodiment with portions broken away;

(7) FIG. 6A is a side perspective schematic view of the top member or the second embodiment;

(8) FIG. 6B is a bottom perspective view of the top member shown in FIG. 6A;

(9) FIG. 6C is a cross-sectional side view of the top member shown in FIG. 6A;

(10) FIG. 6D is a bottom view of the top member shown in FIG. 6A;

- (11) FIG. 7A is a side view of the bottom member of the second embodiment;
(12) FIG. 7B is a side perspective view of the bottom member shown in FIG. 7A;
(13) FIG. 7C is a cross-sectional side view of the bottom member shown in FIG. 7A;
(14) FIG. 7D is a top view of the bottom member shown in FIG. 7A; and,
(15) FIG. 7E is a front perspective view of a post associated with the second embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT(S)

(16) A first embodiment of the dynamic orthotic device of the present invention is depicted in FIGS. 1-4 generally at **20**. Dynamic orthotic device **20** comprises a base **30**, outer rotating disk **40**, inner rotating disk **50** pushbutton **60** and a plurality of massaging balls **70**. The plurality of massaging balls **70** are grouped in two subsets: **70A** and **70B**. Each ball of subset **70A** is received in one of the holes **40A** in outer rotating disk **40** and each of the balls of subset **70B** is received in one of the holes **50B** in inner rotating disk **50**. Although each of the holes **40A** and **50B** are depicted as being larger than the diameter of its respective ball, it is among the features of the present invention that those holes can be smaller than the diameter of the ball while the distance between the support **54** of base **30** beneath outer rotating disk **40** and the disk **40** itself is larger than $\frac{1}{2}$ the diameter of balls **70B** as is the distance between support surface **52** and inner rotating disk **50**. Accordingly, balls **70A** and **70B**, after flexing into the holes via the resilience of the plastic disks **40** and **50**, are retained in the dynamic orthotic device **20** rather than spilling out.

(17) As best seen in FIGS. 3 and 4, pushbutton **60** has an inverted U shape that accommodates spring **62** which, in turn biases pushbutton **60** upwardly. Outer gear-toothed member **64** is attached to outer rotating disk **40** while inner gear-toothed member **66** is attached to inner rotating disk **50**. Two spring-biased arms **68A** protrude downwardly from the outer wall **61** of U-shaped pushbutton **60**, while two additional spring-biased arms **68B** extend downwardly from the inner wall **63** of U-shaped pushbutton **60**. It is to be noted the two arms **68A** extend in the same direction and the two arms **68B** extend in the same (but opposite) direction.

(18) When the dynamic orthotic device **20** is inserted in a shoe, the heel of the wearer will compress pushbutton **60** collapsing spring-biased arms **68A** and **68B** effectively extending them in opposite directions. The tines **69A** and **69B** at the ends of arms **68A** and **68B** will engage gear teeth of members **64** and **66** respectively. As the pressure exerted by the wearer's heel is reduced, spring-biased arms **68A** and **68B** will return to their at rest positions such that tines **69A** and **69B** will cause gear-toothed members **64** and **66** to rotate in clockwise and counter-clockwise directions, respectively. Massaging balls **70A** and **70B** will engage the soles of the wearer's feet in different locations with each step, effectively massaging the soles of the wearer's feet.

(19) Second embodiment, the preferred embodiment, of the dynamic orthotic device of the present invention is shown in FIG. 5 generally at **20'**. Device **20'** includes first upper element **70'** and second lower element **80'**. Upper element **70'** has massaging nodules **74'** on the top surface. Upper element **70'** shown in greater detail in FIGS. 6a-6d, is comprised of a first outer cylinder **71'** and a second inner cylinder **73'** (FIG. 6C). Outer cylinder **71'** has flange **72'** protruding from its lower extremity with a series (eight shown) of holes **76'** equally spaced about its periphery. Second inner cylinder **73'** has a plurality of holes **76b'** aligned with holes **76a'**. A plurality of pins **78a'** are inserted through outer holes **76a'** and are press fit into holes **76b'** (FIG. 5). A recess **75'** is formed in the central top portion of upper element **70'** which provides a sliding engagement with the top of post **90a'** (FIG. 5, FIG. 7E). Post **90a'** is press fit into center cylinder **83'** (FIG. 7C) of lower element **80'**. A stack of spring elements **92'** surround post **90a'** and react between rings **91'** and **93'** received in recess **79'** (FIG. 6C) of upper element **70'** and recess **89'** (FIG. 7D) of lower element **80'**, respectively.

(20) Lower element **80'** is depicted in greater detail in FIGS. 7A-7D. Cylindrical recess **82a'** about upper periphery of lower element **80'** receives O-ring **82'** (FIG. 5) to dampen slightly the movement between upper element **70'** and lower element **80'**. Peripheral flange **87'** has a plurality (three shown) of recesses **84'** with two threaded holes each. Similar to upper element **70'**, lower element

80' has an outer cylinder **81'** and an inner cylinder **83'**. Around the inside wall of outer cylinder **81'**, a lower set of teeth **86'** and an upper set of teeth **88'** are attached, the two sets being offset circumferentially about the outer cylinder **81'** for reasons that will be explained shortly. Eight slots **85'** are positioned about the periphery of outer cylinder **81'** through teeth **88'**. Slots **85'** do not extend through the outer wall of cylinder **81'** (see FIG. 7B).

(21) To assemble the dynamic orthotic device **20'** of the second embodiment, the eight pins **78a'** are inserted through the holes **76a'** in flange **72'** of upper element **70'** and press fit into aligned holes **76b'** around the periphery of inner cylinder **73'** (FIG. 6D). After post **90a'** is press fit into inner cylinder **83'** and the stack of springs 'including rings' and 'positioned there over, the pins **78a'** of upper element **70'** are slipped into the slots **85'** in lower element **80'** (FIG. 7B, 7C) to a position between the two sets of teeth **86'** and **88'** and rotated slightly and pressure released so that the pins engage teeth **88'**. Next, the three locking brackets **90'** are secured in recesses **84'** in flange **87'** by securing the two locking screws into the threaded holes positioned in each recess **84'**. The flanges on locking elements **90'** engage over flange **72'** of upper element **70'** and prevent the elements coming apart, while allowing limited relative motion in an axial direction.

(22) In operation, dynamic orthotic device **20'** will be placed in the heel of a shoe. As device **20'** is stepped upon, upper element **70'** collapses against the stack of springs **92'**. Pins **78a'** will engage the forward faces of teeth **86'** causing a clockwise rotation of roughly 7.5°. As the walker's foot lifts pressure off of upper element **70'**, pins **78a'** initially ride up the vertical face of teeth **86'** and then engage the forward-directed faces of teeth **88'** causing another rotation of roughly 7.5°. The nodules **74'** will massage the bottom of the wearer's foot, repeatedly moving the points of contact, greatly reducing the risk of plantar faciitis and other foot related problems.

(23) Various changes, alternatives, and modifications will become apparent to a person of ordinary skill in the art after a reading of the foregoing specification. It is intended that all such changes, alternatives, and modifications as fall within the scope of the appended claims be considered part of the present invention.

Claims

1. A dynamic orthotic device for massaging a heel portion of the sole of a wearer's foot, said dynamic orthotic device comprising: a) a first upper member comprising a first outer cylinder having a flange protruding outwardly from a lower edge thereof and a second inner cylinder, said first outer cylinder and said second inner cylinder being interconnected by an upper surface; b) a second lower member supporting the first upper member and including a first outer cylinder having an outwardly protruding flange extending outwardly from a lower edge portion of said first outer cylinder and a second inner cylinder, said first and second cylinders of said second lower member being interconnected by a bottom surface and interdigitating with said first and second cylinders of said first upper member, said second lower member further comprising a first set of upwardly directed, forwardly slanting teeth opposed by a second offset set of downwardly directed forwardly slanting teeth, each secured to an inner wall of said first outer cylinder of said second lower member; c) spring means to bias said first member away from said second member; d) rotation means for turning said first member relative to said second member in response to said upper member being weighted and unweighted by the wearer's foot.

2. The dynamic orthotic device of claim 1 wherein said rotation means for turning said first upper member relative to said second lower member comprises a plurality of pins extending outwardly from a second inner wall of said first upper member which engage said first and second sets of forwardly slanting teeth secured to said inner wall of said first outer cylinder of said second lower member, whereby when said upper surface of said first upper member is engaged by a heel portion of the sole of a wearer's foot, said upper member collapses against a force of said spring means causing said plurality of pins to engage said upwardly directed forwardly slanting teeth of said

second lower member and said plurality of pins to engage said downwardly directed forwardly slanting teeth when the heel portion of the sole of the wearer's foot is lifted from said upper surface of said upper member.

3. The dynamic orthotic device of claim 1 wherein said upper surface of said first upper member further comprises a plurality of massaging nodules secured to said upper surface.

4. The dynamic orthotic device of claim 1 further comprising a cylindrical post press fit into and extending upwardly from said inner second cylinder of said second lower member and wherein said spring means comprises a stack of spring elements surrounding said post.

5. A dynamic orthotic device for massaging a heel portion of the sole of a wearer's foot, said dynamic orthotic device comprising: a) a first upper member comprising a first outer cylinder having a flange protruding outwardly from a lower edge thereof and a second inner cylinder, said first outer cylinder and said second inner cylinder being interconnected by an upper surface, said upper surface of said upper member having a plurality of massaging nodules thereon; b) a second lower member supporting the first upper member and comprising a cylinder having an outwardly protruding flange extending outwardly from a lower edge portion of said cylinder, said cylinder interdigitating with said first and second cylinders of said first upper member, said second lower member further comprising a first set of upwardly directed, forwardly slanting teeth opposed by a second offset set of downwardly directed forwardly slanting teeth, each secured to an inner wall of said cylinder of said second lower member; c) spring means to bias said first member away from said second member; d) rotation means for turning said first member relative to said second member in response to said upper member being weighted and unweighted by the wearer's foot.

6. The dynamic orthotic device of claim 5 wherein said rotation means for turning said first upper member relative to said second lower member comprises a plurality of pins extending outwardly from a second inner wall of said first upper member which engage said first and second sets of forwardly slanting teeth secured to said inner wall of said cylinder of said second lower member, whereby when said upper surface of said first upper member is engaged by a heel portion of the sole of a wearer's foot, said upper member collapses against a force of said spring means causing said plurality of pins to engage said upwardly directed forwardly slanting teeth of said second lower member and said plurality of pins to engage said downwardly directed forwardly slanting teeth when the heel portion of the sole of the wearer's foot is lifted from said upper surface of said upper member.
